

Measures of Central Tendency

Class 8-10 Study Notes - Statistics Playlist

1. What is Central Tendency?

Definition: Central Tendency refers to the measure used to determine the center of the distribution of data.

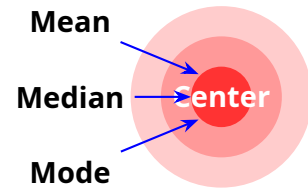
Hello students! Today we are learning one of the most foundational topics in Statistics: **Measures of Central Tendency**. Imagine you have a massive list of numbers. Instead of reading all the numbers, what if you could find just *ONE single number* that represents the center or the "average summary" of all that data? That is exactly what Central Tendency does!

To find this central value, we use three main super-tools: **Mean, Median, and Mode**.

Practice Question:

Q: If a teacher wants to know the overall performance of the class in a test using just one number, what statistical concept is she looking for?

A: Measure of Central Tendency!



Topic Summary: Central Tendency

It is the mathematical way to find the single central or typical value of a dataset using Mean, Median, or Mode.

2. The Mean (Average)

The **Mean** is simply the mathematical average. You add all the numbers together and divide by how many numbers there are.

In Statistics, we have different formulas depending on whether we are calculating the mean for the *entire Population* or just a *Sample* (a small part of the population). The formulas are globally used in data science!

Important Note & Extra Info:

Formulas to Remember:

1. Population Mean (μ - pronounced 'mu'):

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

(Here, N is the total size of the Population data.)

2. Sample Mean ($\bar{x}$ - pronounced 'x-bar'):

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

(Here, n is the total size of the Sample data.)

Real-Life Example:

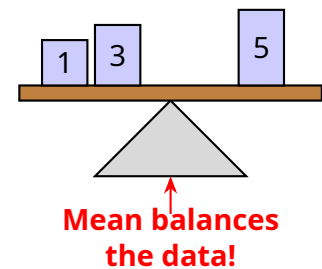
Let's calculate the mean of this dataset: 1, 2, 2, 3, 4, 5

Step 1 (Sum): $1 + 2 + 2 + 3 + 4 + 5 = 17$

Step 2 (Divide): Count is 6. So, $17 \div 6 = 2.83$

Result: The Mean is **2.83**. This number represents the "center" of our data!

μ = Population $\bar{x}$ = Sample



Topic Summary: Mean

The Mean is the average. Sum of all elements divided by the total number of elements. Symbol for Population is μ , and for Sample is $\bar{x}$.

3. The Median (Middle Value)

The problem with Mean: Let's take our previous data (1, 2, 2, 3, 4, 5). The mean was 2.83. Now, what if we add an **Outlier** (a very huge or very tiny number) like 100? The new sum is 117. The new mean is $117 \div 7 = 16.71$. The mean jumped drastically! It no longer represents the real "center" because of one massive number.

To fix this, we use the **Median**. The Median is the exact middle number, and it is *not affected by outliers*!

Important Note & Extra Info:

Steps to find the Median:

1. **Sort** the data in increasing order (smallest to largest).
2. Find the **central element**.
 - If the count is **Odd**, pick the exact middle number.
 - If the count is **Even**, take the average of the two middle numbers.

Real-Life Example:

Example 1 (Odd Length): Data = 1, 2, 3, 4, 5.

The exact middle number is **3**. So, Median = 3.

Example 2 (Even Length with Outliers!): Data = 1, 2, 2, 3, 4, 5, 100, 101.

Count is 8 (Even). The middle two numbers are 3 and 4.

Median = $(3 + 4) \div 2 = 3.5$.

Notice how the massive 100 and 101 did not ruin our center! The median safely stayed at 3.5.

Sort First!
1 2 3 4 5 →

Even Length

(2) (3) (4) (5)

↓
Avg = 3.5
(Median)

Topic Summary: Median

The Median is the exact middle value after sorting the data. It is the best tool to use when your data has extreme values (Outliers).

4. The Mode (Most Frequent)

The **Mode** is the easiest to find! It is simply the element that appears the most number of times in your dataset (the most frequent element).

Real-Life Example:

Let's look at this data: 1, 2, 2, 3, 4, 5, 5, 5, 5

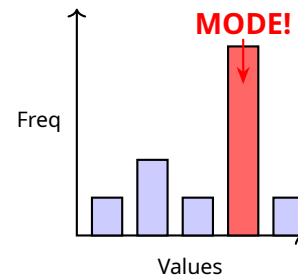
- Number 2 is repeated 2 times.
- Number 5 is repeated **4 times**.

Because 5 appears the most frequently, the **Mode = 5**. That becomes our measure of central tendency!

Practice Question:

Q: A shoe store sells the following shoe sizes in a day: 7, 8, 8, 9, 8, 10. What is the mode?

A: The mode is 8, because it was sold the most times!



5. When to use Mean, Median, or Mode?

In Data Science, knowing *when* to use which formula is the most important skill! If you have missing data in your dataset, here is how you decide which central tendency to use to fill in the blanks:

Use MEAN when:

Your data is **Normally Distributed** and has NO massive outliers.
Example: Filling missing average heights of students.

Use MEDIAN when:

Your data has **Outliers** (extreme high or low values) that would ruin the average.
Example: Filling missing ages when one age is 100.

Use MODE when:

You are dealing with **Categorical Features** (text labels) instead of numbers.
Example: Filling a missing Gender with the most frequent one (Male/Female).

Overall Revision Summary

- **Central Tendency:** A single number that represents the center point of a dataset.
- **Mean:** The calculated average. Formula depends on if it is a Population (μ) or Sample ($\bar{x}$). Used for normal data.
- **Outliers:** Extremely large or extremely small numbers that distort the Mean.
- **Median:** The middle number of a sorted list. It is the best defense against Outliers! If the list is even, average the two middle numbers.
- **Mode:** The element that occurs the most frequently. Perfect for Categorical data (like finding the most common Gender or Color).

Keep practicing! Building a strong foundation in these three concepts is the key to mastering Data Science!